

APRIL 2025

2504225 / UMSA21I

Time: Three Hours

Maximum Marks: 75

SECTION - A ($10 \times 2 = 20$ Marks)

Answer Any TEN Questions

1. Define Bernoulli's formula.
2. Find the value of $\int_0^{\pi/2} \sin^7 x \cos^5 x \, dx$.
3. Define an even function.
4. Write the expression of the Fourier coefficient of a_0 .
5. Solve $\frac{d^2 y}{dx^2} - 4y = 6e^{5x}$.
6. Solve $p + q = 1$.
7. Define Laplace transforms.
8. Find $L(e^{2t})$.
9. Find $L(\cosh 7t)$.
10. Define gradient of the scalar function.
11. Find $\text{grad } f$ if $f = x^2 + y^2$ and determine its magnitude at (3,4).
12. If $\vec{F} = x^2\vec{i} + y^2\vec{j} + z^2\vec{k}$ then find $\nabla \times \vec{F}$.

SECTION - B ($5 \times 5 = 25$ Marks)

Answer Any FIVE Questions

13. Evaluate $\int x^4 e^{2x} \, dx$.
14. Evaluate $\int \sin^6 x \, dx$.
15. Find the Fourier series for $f(x) = \begin{cases} -K, & -\pi < x < 0 \\ K, & 0 < x < \pi \end{cases}$, hence deduce that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$.
16. Solve $(D^2 - 4D - 5)y = \cos x + e^{-x}$.
17. Find the singular integral of $z = px + qy + pq$.
18. Find $L(\sin^2 t \, 2t)$.
19. Find the directional derivative of $f = xyz$ at (1,1,1) in the direction of $\vec{i} + \vec{j} + \vec{k}$.

SECTION - C ($3 \times 10 = 30$ Marks)

Answer Any THREE Questions

20. Evaluate $\int \cos^7 x \, dx$ using reduction formula.
21. Express $f(x) = x \sin x$ as a Fourier series in $0 \leq x \leq 2\pi$.
22. Solve $(D^2 + 3D + 2)y = \sin 3x \cos 2x$.
23. Find $L[\sin^2 t \cos^3 t]$.
24. If $\nabla\phi = (y^2 - 2xyz^3)\vec{i} + (3 + 2xy - x^2z^3)\vec{j} + (6z^3 - 3x^2yz^2)\vec{k}$ find ϕ .